

MATHEMATICS

Chapter 12: LIMITS AND DERIVATIVES



LIMITS AND DERIVATIVES

Some useful results

1. $(a^2 - b^2) = (a + b)(a - b)$
2. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
3. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
4. $a^4 - b^4 = (a^2 - b^2)(a^2 + b^2) = (a + b)(a - b)(a^2 + b^2)$
5. $(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$
6. $\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} \dots$
7. $\log(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} \dots$
8. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$
9. $e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$
10. $a^x = 1 + x(\log a) + \frac{x^2}{2!}(\log_e a)^2 + \dots$
11. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$
12. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$
13. $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$
14. $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
15. $\cos(A \pm B) = \cos A \cos B \pm \sin A \sin B$
16. $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$
17. $\tan A \pm \tan B = \frac{\sin(A \pm B)}{\cos A \cos B}$
18. $\tan A - \tan B = \tan(A - B)\{1 + \tan A \tan B\}$
19. $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$
20. $\sin C - \sin D = 2 \sin \frac{C-D}{2} \cos \frac{C+D}{2}$

$$21. \cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$22. \cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$$

$$23. 2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$24. 2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$25. 2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

Key Concepts

1. The expected value of the function as dictated by the points to the left of a point defines the left-hand limit of the function at that point. The limit $\lim_{x \rightarrow a^-} f(x)$ is the expected value of f at $x = a$ given the values of f near x to the left of a .
2. The expected value of the function as dictated by the points to the right of point a defines the right-hand limit of the function at that point. The limit $\lim_{x \rightarrow a^+} f(x)$ is the expected value of f at $x = a$ given the values of f near x to the right of a .
3. Let $y = f(x)$ be a function. Suppose that a and L are numbers such that as x gets closer and closer to a , $f(x)$ gets closer and closer to L we say that the limit of $f(x)$ at $x = a$ is L , i.e., $\lim_{x \rightarrow a} f(x) = L$.
4. Limit of a function at a point is the common value of the left- and right-hand limit if they coincide, i.e., $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$.
5. Real life examples of LHL and RHL
 - a) If a car starts from rest and accelerates to 60 km/hr in 8 seconds, which means the initial speed of the car is 0 and it reaches 60 km 8 seconds after the start. On recording the speed of the car, we can see that this sequence of numbers is approaching 60 km in such a way that each member of the sequence is less than 60. This sequence illustrates the concept of approaching a number from the left of that number.
 - b) Boiled milk which is at a temperature of 100 degrees is placed on a shelf; temperature goes on dropping till it reaches room temperature.
As the time duration increases, temperature of milk, t , approaches room temperature say 30° . This sequence illustrates the concept of approaching a number from the right of that number.

6. Let f and g be two functions such that both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exists then

a. Limit of the sum of two functions is the sum of the limits of the functions,

$$\text{i.e. } \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x).$$

b. Limit of the difference of two functions is the difference of the limits of the functions,

$$\text{i.e. } \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x).$$

c. Limit of the product of two functions is the product of the limits of the functions,

$$\text{i.e. } \lim_{x \rightarrow a} [f(x).g(x)] = \lim_{x \rightarrow a} f(x). \lim_{x \rightarrow a} g(x).$$

d. Limit of the quotient of two functions is the quotient of the limits of the functions (whenever the denominator is non zero),

$$\text{i.e. } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

e. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ exists, then

$$\lim_{x \rightarrow a} \left| 1 - \frac{f(x)}{g(x)} \right| = e^{\lim_{x \rightarrow a} \frac{f(x)}{g(x)}}$$

f. If $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} g(x) = \infty$ such that

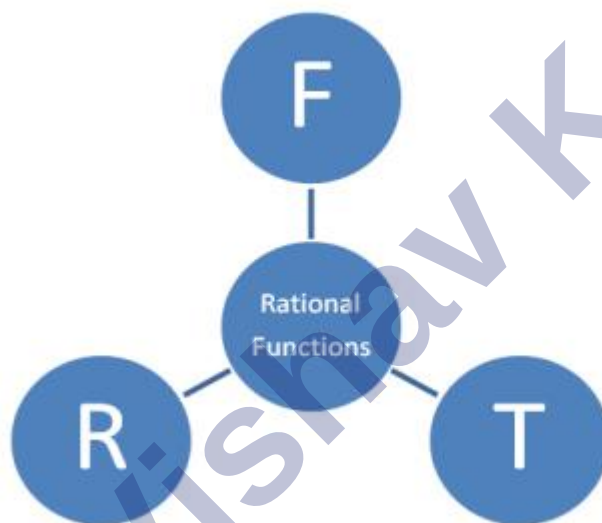
$$\lim_{x \rightarrow a} |f(x) - 1|g(x) \text{ exists, then,}$$

$$\lim_{x \rightarrow a} f(x)^{g(x)} = e^{\lim_{x \rightarrow a} |f(x) - 1|g(x)}$$

7. For any positive integer n ,

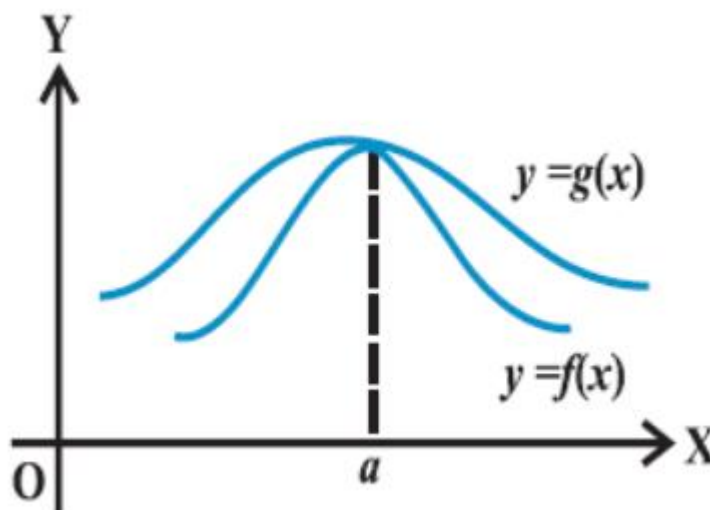
$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$$

8. Limit of a polynomial function can be computed using substitution or algebra of limits.
9. The following methods are used to evaluate algebraic limits:
- Direct substitution method
 - Factorization method
 - Rationalization method
 - By using some standard limits
 - Method of evaluation of algebraic limits at infinity
10. For computing the limit of a rational function when direct substitution fails, use factorisation, rationalisation or the theorem.

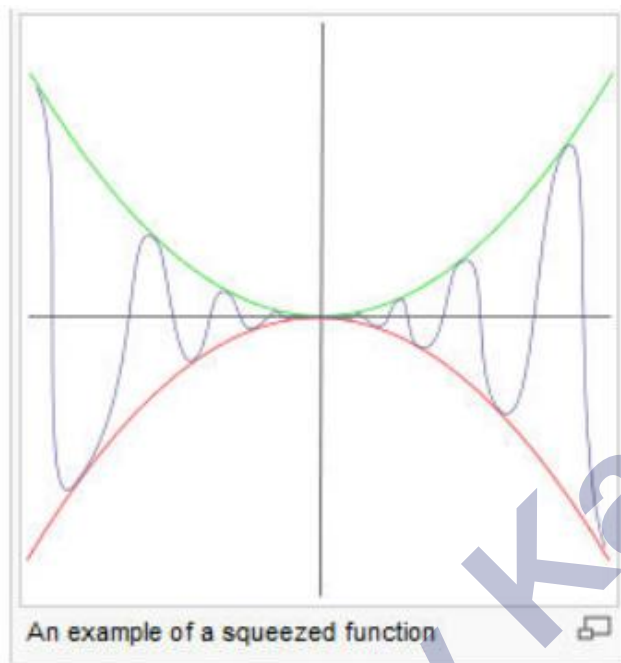


11. Let f and g be two real valued functions with the same domain such that $f(x) \leq g(x)$ for all x in the domain of definition. For some a , if both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist, then

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x).$$



Let f , g and h be real functions such that $f(x) \leq g(x) \leq h(x)$ for all x in the common domain of definition. For some real number a , if $\lim_{x \rightarrow a} f(x) = \ell = \lim_{x \rightarrow a} h(x)$ then $\lim_{x \rightarrow a} g(x) = \ell$



12. Suppose f is a real valued function and a is a point in its domain of definition. The derivative of f at a is defined by

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Provided this limit exists and is finite. Derivative of $f(x)$ at a is denoted by $f'(a)$.

13. A function is differentiable in its domain if it is always possible to draw a unique tangent at every point on the curve.

14. Finding the derivative of a function using definition of derivative is known as the first principle of derivatives or ab-initio method.

15. Differentiation of a constant function is zero.

16. If $f(x)$ is a differentiable function and ' c ' is a constant, then $\frac{d}{dx}(c \cdot f(x)) = c \cdot \frac{d}{dx}f(x)$.

17. Let f and g be two functions such that their derivatives are defined in a common domain. Then

i. Derivative of the sum of two functions is the sum of the derivatives of the functions.

$$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$$

ii. Derivative of the difference of two functions is the difference of the derivatives of the functions.

$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}f(x) - \frac{d}{dx}g(x)$$

iii. Derivative of the product of two functions is given by the following products rule.

$$\frac{d}{dx}[f(x) \cdot g(x)] = \frac{d}{dx}f(x) \cdot g(x) + f(x) \cdot \frac{d}{dx}g(x)$$

iv. Derivative of quotient of two functions is given by the following quotient rule (whenever the denominator is non-zero).

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{\frac{d}{dx}f(x) \cdot g(x) - f(x) \cdot \frac{d}{dx}g(x)}{(g(x))^2}$$

18. Generalization of the product rule: Let $f(x)$, $g(x)$ and $h(x)$ be three differentiable functions.

Then

$$\begin{aligned} & \frac{d}{dx}[f(x) \cdot g(x) \cdot h(x)] \\ &= \frac{d}{dx}[f(x)] \cdot g(x) \cdot h(x) + f(x) \cdot \frac{d}{dx}[g(x)] \cdot h(x) + f(x) \cdot g(x) \cdot \frac{d}{dx}[h(x)] \end{aligned}$$

19. Derivative of $f(x) = x^n$ is nx^{n-1} for any positive integer n .

20. Let $f(x) = a_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \dots + 2a_2x + a_1$.

Now, a_2x are all real numbers and $a_n \neq 0$. Then, the derivative function is given by

$$\frac{df(x)}{dx} = na_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \dots + 2a_2x + a_1.$$

21. For a function f and a real number a , $\lim_{x \rightarrow a} f(x)$ and $f(a)$ may not be same (In fact, one may be defined and not the other one).

22. Standard derivatives

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
x^n	nx^{n-1}
c	0
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$
e^x	e^x
$\frac{1}{\sqrt{x}}$	$-\frac{1}{2}x^{-\frac{3}{2}}$
$\frac{1}{x}$	$-\frac{1}{x^2}$
a^x	$a^x \log_e a$
$\log_e x$	$\frac{1}{x}$

23. The derivative is the instantaneous rate of change in the terms of Physics and is the slope of the tangent at a point.
24. A function is not differentiable at the points where it is not defined or at the points where the unique tangent cannot be drawn.
25. Consider that $f'(x)$, $\frac{dy}{dx}$, $\frac{df(x)}{dx}$ and y' are all different notations for the derivative with respect to x .

26. Key Formulae

1. $\lim_{x \rightarrow \infty} c = c$
2. $\lim_{x \rightarrow -\infty} c = c$
3. $\lim_{x \rightarrow \infty} \frac{c}{x^n} = 0, n > 0$
4. $\lim_{x \rightarrow -\infty} \frac{c}{x^n} = 0, n \in \mathbb{N}$
5. $\lim_{x \rightarrow +\infty} x \rightarrow +\infty$
6. $\lim_{x \rightarrow -\infty} x \rightarrow -\infty$
7. $\lim_{x \rightarrow +\infty} x^2 \rightarrow +\infty$
8. $\lim_{x \rightarrow -\infty} x^2 \rightarrow \infty$
9. $\lim_{x \rightarrow \infty} e^x \rightarrow \infty$
10. $\lim_{x \rightarrow -\infty} e^{-x} \rightarrow \infty$
11. $\lim_{x \rightarrow \infty} e^{-x} \rightarrow 0$
12. $\lim_{x \rightarrow -\infty} e^x \rightarrow 0$
13. $\lim_{x \rightarrow \infty} a^x \rightarrow 0$, if $|a| < 1$
14. $\lim_{x \rightarrow \infty} a^x \rightarrow \infty$, if $|a| > 1$
15. $\lim_{x \rightarrow 0^+} \log_a x \rightarrow -\infty$ and $\lim_{x \rightarrow \infty} \log_a x \rightarrow \infty$, where $a > 1$
16. $\lim_{x \rightarrow 0^+} \log_a x \rightarrow \infty$ and $\lim_{x \rightarrow \infty} \log_a x \rightarrow -\infty$, where $0 < a < 1$
17. $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$
18. $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$
19. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \log_e e = 1$
20. $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$
21. $\lim_{x \rightarrow 0} \left(1 + \frac{1}{x}\right)^x = e$
22. $\lim_{x \rightarrow 0} (1+\lambda x)^{\frac{1}{x}} = e^\lambda$
23. $\lim_{x \rightarrow 0} \left(1 + \frac{\lambda}{x}\right)^x = e^\lambda$

$$24. \lim_{x \rightarrow 0} \sin x = 0$$

$$25. \lim_{x \rightarrow 0} \cos x = 1$$

$$26. \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$27. \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

$$28. \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$29. \lim_{x \rightarrow a} \frac{\sin(x - a)}{(x - a)} = 1$$

$$30. \lim_{x \rightarrow a} \frac{\tan(x - a)}{(x - a)} = 1$$

Steps for finding the left-hand limit

- Step 1:** Get the function $\lim_{x \rightarrow a^-} f(x)$
- Step 2:** Substitute $x = a - h$ and replace $x \rightarrow a^-$ by $h \rightarrow 0$ to get $\lim_{h \rightarrow 0} f(a - h)$
- Step 3:** Using appropriate formula simplify the given function.
- Step 4:** The final value is the left-hand limit of the function at $x = a$.

Steps for finding the right-hand limit

- Step 1:** Get the function $\lim_{x \rightarrow a^+} f(x)$
- Step 2:** Substitute $x = a + h$ and replace $x \rightarrow a^+$ by $h \rightarrow 0$ to get $\lim_{h \rightarrow 0} f(a + h)$
- Step 3:** Using appropriate formula simplify the given function.
- Step 4:** The final value is the right-hand limit of the function at $x = a$.

Steps for factorisation method

- Step 1:** Get the function $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$, where $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$
- Step 2:** Factorize $f(x)$ and $g(x)$.
- Step 3:** Cancel out the common factors.

4. **Step 4:** Use the direct substitution method to find the final limit.

Steps for rationalisation method

1. When the numerator or denominator or both involve expression takes the form $\frac{0}{0}, \frac{\infty}{\infty}$ we can use this method.

In this method, factor out the numerator and the denominator separately and cancel the common factor

Example:

Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{2+x} - \sqrt{2}}{x}$:

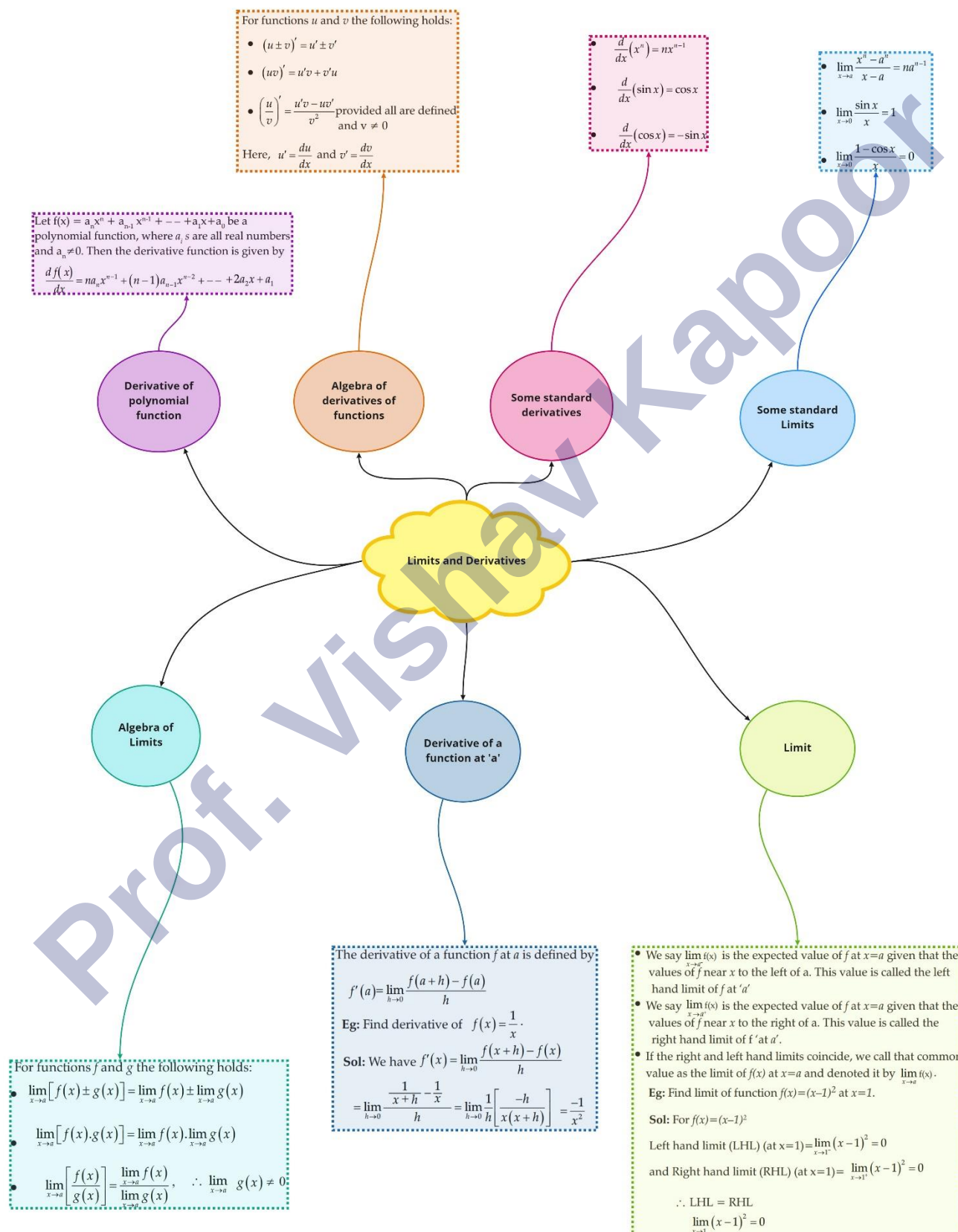
Solution:

At $x=0$, $\frac{\sqrt{2+x} - \sqrt{2}}{x} \rightarrow \frac{0}{0}$

Thus, rationalising the numerator, we have,

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{2+x} - \sqrt{2}}{x} &= \lim_{x \rightarrow 0} \frac{(\sqrt{2+x} - \sqrt{2})(\sqrt{2+x} + \sqrt{2})}{x(\sqrt{2+x} + \sqrt{2})} \\ &= \lim_{x \rightarrow 0} \frac{2+x-2}{x(\sqrt{2+x} + \sqrt{2})} \\ &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{2+x} + \sqrt{2}} \\ &= \frac{1}{2\sqrt{2}} \end{aligned}$$

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Important Questions

Multiple Choice questions-

Question 1. The expansion of $\log(1 - x)$ is:

- (a) $x - x^2/2 + x^3/3 - \dots$
- (b) $x + x^2/2 + x^3/3 + \dots$
- (c) $-x + x^2/2 - x^3/3 + \dots$
- (d) $-x - x^2/2 - x^3/3 - \dots$

Question 2. The value of $\lim_{x \rightarrow a} (a \times \sin x - x \times \sin a)/(ax^2 - xa^2)$ is

- (a) $= (a \times \cos a + \sin a)/a^2$
- (b) $= (a \times \cos a - \sin a)/a^2$
- (c) $= (a \times \cos a + \sin a)/a$
- (d) $= (a \times \cos a - \sin a)/a$

Question 3. $\lim_{x \rightarrow -1} [1 + x + x^2 + \dots + x^{10}]$ is

- (a) 0
- (b) 1
- (c) -1
- (d) 2

Question 4. The value of the limit $\lim_{x \rightarrow 0} \{\log(1 + ax)\}/x$ is

- (a) 0
- (b) 1
- (c) a
- (d) $1/a$

Question 5. The value of the limit $\lim_{x \rightarrow 0} (\cos x)\cot^{2x}$ is

- (a) 1
- (b) e
- (c) $e^{1/2}$
- (d) $e^{-1/2}$

Question 6. Then value of $\lim_{x \rightarrow 1} (1 + \log x - x)/(1 - 2x + x^2)$ is

- (a) 0

(b) 1

(c) $1/2$ (d) $-1/2$ Question 7. The value of $\lim_{y \rightarrow 0} \{(x + y) \times \sec(x + y) - x \times \sec x\}/y$ is(a) $x \times \tan x \times \sec x$ (b) $x \times \tan x \times \sec x + x \times \sec x$ (c) $\tan x \times \sec x + \sec x$ (d) $x \times \tan x \times \sec x + \sec x$ Question 8. $\lim_{x \rightarrow 0} (e^{x^2} - \cos x)/x^2$ is equals to

(a) 0

(b) 1

(c) $2/3$ (d) $3/2$ Question 9. The expansion of a^x is:(a) $a^x = 1 + x/1! \times (\log a) + x^2/2! \times (\log a)^2 + x^3/3! \times (\log a)^3 + \dots$ (b) $a^x = 1 - x/1! \times (\log a) + x^2/2! \times (\log a)^2 - x^3/3! \times (\log a)^3 + \dots$ (c) $a^x = 1 + x/1 \times (\log a) + x^2/2 \times (\log a)^2 + x^3/3 \times (\log a)^3 + \dots$ (d) $a^x = 1 - x/1 \times (\log a) + x^2/2 \times (\log a)^2 - x^3/3 \times (\log a)^3 + \dots$ Question 10. The value of the limit $\lim_{n \rightarrow 0} (1 + an)^{b/n}$ is:(a) e^a (b) e^b (c) e^{ab} (d) $e^{a/b}$ **Very Short Questions:**1. Evaluate $\lim_{x \rightarrow 3} \left[\frac{x^2 - 9}{x - 3} \right]$ 2. Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$ 3. Find derivative of $2x$.4. Find derivative of $\sqrt{\sin 2x}$

5. Evaluate $\lim_{x \rightarrow 0} \frac{\sin^2 4x}{x^2}$
6. What is the value of $\lim_{x \rightarrow a} \left(\frac{x^2 - a^n}{x - a} \right)$
7. Differentiate $\frac{2x}{x}$
8. If $y = e^{\sin x}$ Find $\frac{dy}{dx}$
9. Evaluate $\lim_{x \rightarrow 1} \frac{x^{15} - 1}{x^{10} - 1}$
10. Differentiate $x \sin x$ with respect to x .

Short Questions:

1. Prove that $\lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right) = 1$
2. Evaluate $\lim_{x \rightarrow 1} \frac{(2x-3)(\sqrt{x}-1)}{(2x^2+x-3)} = 1$
3. Evaluate $\lim_{x \rightarrow 0} \frac{x \tan 4x}{1 - \cos 4x}$
4. If $y = \frac{(1-\tan x)}{(1+\tan x)}$. Show that $\frac{dy}{dx} = \frac{-2}{(1+\sin 2x)}$
5. Differentiate $e^{\sqrt{\cot x}}$

Long Questions:

1. Differentiate $\tan x$ from first principle.
2. Differentiate $(x + 4)^6$ From first principle.
3. Find derivative of $\operatorname{cosec} x$ by first principle.
4. Find the derivatives of the following functions:

$$(i) \left(x - \frac{1}{x} \right)^3 \quad (ii) \frac{(3x+1)(2\sqrt{x-1})}{\sqrt{x}}$$

5. Find the derivative of $\sin(x + 1)$ with respect to x from first principle.

Assertion Reason Questions:

1. In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements, mark the correct answer as.

Assertion (A) $\lim_{x \rightarrow 1} \frac{ax^2 + bx + c}{cx^2 + bx + a}$ is

equal to 1, where $a + b + c \neq 0$.

Reason (R) $\lim_{x \rightarrow -2} \frac{\frac{1}{x} + \frac{1}{2}}{x + 2}$ is equal to $\frac{1}{4}$.

- (i) Both assertion and reason are true and reason is the correct explanation of assertion.
 - (ii) Both assertion and reason are true but reason is not the correct explanation of assertion.
 - (iii) Assertion is true but reason is false.
 - (iv) Assertion is false but reason is true.
2. In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements, mark the correct answer as.

Assertion (A) $\lim_{x \rightarrow 0} \frac{\sin ax}{bx}$ is equal to $\frac{a}{b}$.

Reason (R) $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

- (i) Both assertion and reason are true and reason is the correct explanation of assertion.
- (ii) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (iii) Assertion is true but reason is false.
- (iv) Assertion is false but reason is true.

Answer Key:

MCQ:

1. (d) $-x - x^2/2 - x^3/3 - \dots$
2. (b) $= (a \times \cos a - \sin a)/a^2$
3. (b) 1
4. (c) a
5. (d) $e^{-1/2}$

6. (d) $-1/2$

7. (d) $x \times \tan x \times \sec x + \sec x$

8. (d) $3/2$

9. (a) $a^x = 1 + x/1! \times (\log a) + x^2/2! \times (\log a)^2 + x^3/3! \times (\log a)^3 + \dots$

10.(c) e^{ab}

Very Short Answer:

1.

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} \frac{0}{0} \text{ form}$$

$$\lim_{x \rightarrow 3} \frac{(x+3)(\cancel{x-3})}{(\cancel{x-3})} = 3+3 = 6$$

2.

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$$

$$= \lim_{3x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{3}{5}$$

$$= 1 \times \frac{3}{5} = \frac{3}{5} \left[\because \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right]$$

3. Let $y = 2^x$

$$\frac{dy}{dx} = \frac{d}{dx} 2 = 2^x \log 2$$

4.

$$\frac{d}{dx} \sqrt{\sin 2x} = \frac{1}{2\sqrt{\sin 2x}} \frac{d}{dx} \sin 2x$$

$$= \frac{1}{2\sqrt{\sin 2x}} \times 2 \cos 2x$$

$$= \frac{\cos 2x}{\sqrt{\sin 2x}}$$

5.

$$\lim_{x \rightarrow 0} \frac{\sin^2 4x}{x^2 4^2} \times 4^2 = \lim_{4x \rightarrow 0} \left(\frac{\sin 4x}{4x} \right)^2 \times 16$$

$$= 1 \times 16 = 16$$

6.

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = 1$$

7.

$$\begin{aligned} \frac{d}{dx} \frac{2^x}{x} &= \frac{x \frac{d}{dx} 2^x - 2^x \frac{d}{dx} x}{x^2} \\ &= \frac{x \times 2^x \log 2 - 2^x \times 1}{x^2} \\ &= 2^x \frac{[x + \log 2 - 1]}{x^2} \end{aligned}$$

8.

$$\begin{aligned} y &= e^{\sin x} \\ \frac{dy}{dx} &= \frac{d}{dx} e^{\sin x} \\ &= e^{\sin x} \times \cos x = \cos x e^{\sin x} \end{aligned}$$

9.

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{x^{15} - 1}{x^{10} - 1} \\ &= \frac{\lim_{x \rightarrow 1} \frac{x^{15} - 1^{15}}{x - 1}}{\lim_{x \rightarrow 1} \frac{x^{10} - 1^{10}}{x - 1}} = \frac{15 \times 1^{14}}{10 \times 1^9} \\ &= \frac{15}{10} = \frac{3}{2} \end{aligned}$$

10.

$$\begin{aligned} \frac{d}{dx} x \sin x &= x \cos x + \sin x \cdot 1 \\ &= x \cos x + \sin x \end{aligned}$$

Short Answer:

1. We have

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

$$\lim_{x \rightarrow 0} \left\{ \frac{\left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right] - 1}{x} \right\} \left[\because e^x = 1 + x + \frac{x^2}{2!} + \dots \right]$$

$$\lim_{x \rightarrow 0} \left\{ \frac{x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots}{x} \right\}$$

$$\lim_{x \rightarrow 0} \left\{ \frac{1 + \frac{x}{2!} + \frac{x^2}{3!} + \dots}{1} \right\}$$

$$= 1 + 0 = 1$$

2.

$$\lim_{x \rightarrow 1} \frac{(2x-3)(\sqrt{x}-1)}{(2x^2+x-3)}$$

$$= \lim_{x \rightarrow 1} \frac{(2x-3)(\sqrt{x}-1)}{(2x+3)(x-1)}$$

$$\lim_{x \rightarrow 1} \frac{(2x-3)(\sqrt{x}-1)}{(2x+3)(x-1)} \times \frac{(\sqrt{x}+1)}{(\sqrt{x}+1)}$$

$$\lim_{x \rightarrow 1} \frac{(2x-3)(\cancel{x-1})}{(2x+3)(\cancel{x-1})(\sqrt{x}+1)}$$

$$\lim_{x \rightarrow 1} \frac{(2x-3)}{(2x+3)(\sqrt{x}+1)} = \frac{2 \times 1 - 3}{(2 \times 1 + 3)(\sqrt{1} + 1)}$$

$$= \frac{-1}{10}$$

3.

$$\lim_{x \rightarrow 0} \frac{x \tan 4x}{1 - \cos 4x}$$

$$= \lim_{x \rightarrow 0} \frac{x \sin 4x}{\cos 4x [2 \sin^2 2x]}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{2x \cancel{\sin 2x} \cos 2x}{\cos 4x (2 \sin^2 2x)} \\
 &= \lim_{x \rightarrow 0} \left(\frac{\cos 2x}{\cos 4x} \cdot \frac{2x}{\sin 2x} \times \frac{1}{2} \right) \\
 &= \frac{1}{2} \lim_{2x \rightarrow 0} \frac{\cos 2x}{\cos 4x} \times \lim_{2x \rightarrow 0} \left(\frac{2x}{\sin 2x} \right) = \frac{1}{2} \times 1 = \frac{1}{2}
 \end{aligned}$$

4.

$$\begin{aligned}
 y &= \frac{(1 - \tan x)}{(1 + \tan x)} \\
 \frac{dy}{dx} &= \frac{(1 + \tan x) \frac{d}{dx}(1 - \tan x) - (1 - \tan x) \frac{d}{dx}(1 + \tan x)}{(1 + \tan x)^2} \\
 &= \frac{(1 + \tan x)(-\sec^2 x) - (1 - \tan x)\sec^2 x}{(1 + \tan x)^2} \\
 &= \frac{-\sec^2 x - \cancel{\tan x \sec^2 x} - \sec^2 x + \cancel{\tan x \sec^2 x}}{(1 + \tan x)^2} \\
 &= \frac{-2\sec^2 x}{(1 + \tan x)^2} = \frac{-2}{\cos^2 x \left[1 + \frac{\sin x}{\cos x} \right]^2} \\
 &= \frac{-2}{\cancel{\cos^2 x} \left[\frac{\cos x + \sin x}{\cancel{\cos^2 x}} \right]^2} \\
 &= \frac{-2}{\cos^2 x + \sin^2 x + 2\sin x \cos x} = \frac{-2}{1 + \sin^2 x} \\
 \therefore \frac{dy}{dx} &= \frac{-2}{1 + \sin 2x}
 \end{aligned}$$

Hence Proved.

5.

$$\text{Let } y = e^{\sqrt{\cot x}}$$

$$\frac{dy}{dx} = \frac{d}{dx} e^{\sqrt{\cot x}} = e^{\sqrt{\cot x}} \frac{d}{dx} \sqrt{\cot x}$$

$$= e^{\sqrt{\cot x}} \times \frac{1}{2\sqrt{\cot x}} \cdot \frac{d}{dx} \cot x$$

$$= \frac{e^{\sqrt{\cot x}}}{2\sqrt{\cot x}} - \cos \sec^2 x$$

$$= \frac{-\cos \sec^2 e^{\sqrt{\cot x}}}{2\sqrt{\cot x}}$$

Long Answer:

1.

$$f(x) = \tan x$$

$$f(x+h) = \tan(x+h)$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{\sin(x+h)}{\cos(x+h)} - \frac{\sin x}{\cos x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin(x+h)\cos x - \cos(x+h)\sin x}{h \cos(x+h)\cos x}$$

$$= \lim_{h \rightarrow 0} \frac{\sin[x+h-x]}{h \cos(x+h)\cos x} \left[\because \sin(A-B) = \sin A \cos B - \cos A \sin B \right]$$

$$= \lim_{h \rightarrow 0} \frac{\sinh}{h \cos(x+h)\cos x}$$

$$= \frac{\lim_{h \rightarrow 0} \frac{\sinh}{h}}{\lim_{h \rightarrow 0} \cos(x+h)\cos x} = \frac{1}{\cos x \cdot \cos x} \left[\because \lim_{h \rightarrow 0} \frac{\sinh}{h} = 1 \right]$$

$$= \frac{1}{\cos^2 x} = \sec^2 x$$

2.

$$\text{let } f(x) = (x+4)^6$$

$$f(x+h) = (x+h+4)^6$$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h+4)^6 - (x+4)^6}{h} \\ &= \lim_{(x+h+4) \rightarrow (x+4)} \frac{(x+h+4)^6 - (x+4)^6}{(x+h+4) - (x+4)} \\ &= 6(x+4)^{(6-1)} \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \\ &= 6(x+4)^5 \end{aligned}$$

3.

proof let $f(x) = \operatorname{cosec} x$

$$\begin{aligned} \text{By def, } f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\operatorname{cosec}(x+h) - \operatorname{cosec} x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{1}{\sin(x+h)} - \frac{1}{\sin x}}{h} = \lim_{h \rightarrow 0} \frac{\sin x - \sin(x+h)}{h \sin(x+h) \sin x} \\ &= \lim_{h \rightarrow 0} \frac{2 \cos \frac{x+x+h}{2} \sin \frac{x-x+h}{2}}{h \sin(x+h) \sin x} \\ &= \lim_{h \rightarrow 0} \frac{2 \cos \left(x + \frac{h}{2}\right) \sin \left(-\frac{h}{2}\right)}{h \sin(x+h) \sin x} \\ &= \frac{\lim_{\frac{h}{2} \rightarrow 0} \cos \left(x + \frac{h}{2}\right)}{\cos x} \cdot \lim_{\frac{h}{2} \rightarrow 0} \frac{\sin \frac{h}{2}}{\frac{h}{2}} \\ &= -\frac{\cos x}{\sin x \cdot \sin x} \cdot 1 = -\operatorname{cosec} x \cot x \end{aligned}$$

4.

$$(i) \text{ let } f(x) = \left(x - \frac{1}{x}\right)^3 = x^3 - \frac{1}{x^3} - 3x + \frac{1}{x} \left(x - \frac{1}{x}\right)$$

$$= x^3 - x^{-3} - 3x + 3x^{-1}. \text{ d. ff wr.t. } x, \text{ we get}$$

$$f'(x) = 3 \times x^2 - (-3)x^{-4} - 3 \times 1 + 3 \times (-1)x^{-2}$$

$$= 3x^2 + \frac{3}{x^4} - 3 - \frac{3}{x^2}.$$

$$(ii) \text{ let } f(x) = \frac{(3x+1)(2\sqrt{x}-1)}{\sqrt{x}} = \frac{6x^{\frac{3}{2}} - 3x + 2\sqrt{x} - 1}{\sqrt{x}}$$

$$= 6x - 3x^{\frac{1}{2}} + 2 - x^{-\frac{1}{2}}, \text{ d. ff wr.t. } x, \text{ we get}$$

$$f'(x) = 6 \times 1 - 3 \times \frac{1}{2} \times x^{-\frac{1}{2}} + 0 - \left(-\frac{1}{2}\right)x^{-\frac{3}{2}}$$

$$= 6 - \frac{3}{2\sqrt{x}} + \frac{1}{2x^{\frac{3}{2}}}.$$

5.

$$\text{let } f(x) = \sin(x+1)$$

$$f(x+h) = \sin(x+h+1)$$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin(x+h+1) - \sin(x+1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 \cos \left[\frac{x+h+1+x+1}{2} \right] \sin \left[\frac{x+h+1-x-1}{2} \right]}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 \cos \left[x+1+\frac{h}{2} \right] \sin \frac{h}{2}}{h} \\ &= \lim_{h \rightarrow 0} 2 \cos \left(x+1+\frac{h}{2} \right) \times \lim_{h \rightarrow 0} \frac{\sin \frac{h}{2}}{\frac{h}{2}} \\ &= \cos(x+1) \times 1 = \cos(x+1) \end{aligned}$$

Assertion Reason Answer:

- (iii) Assertion is true but reason is false.
- (i) Both assertion and reason are true and reason is the correct explanation of assertion.